

Early Disease Detection using Morphological Analysis of Cells with Deep Learning

Abstract—With an increase in the number of patients diagnosed with blood-related diseases, early detection of these diseases in today's health care field becomes essential. Blood diseases' detection can be made through conventional medical diagnosis of detecting abnormalities in blood smear images under the microscope. Nonetheless, manual detection through microscopic observation is reliant on the medical professional's skills and therefore consumes too much time and may lead to mistakes that affect patient treatment results. The proposed deep learning framework aims at diagnosing hematological diseases through morphology-based classification of blood cell images. Morphological detection and classification of diseases such as Acute Lymphoblastic Leukemia, malaria, sickle cell anemia, and normal blood cells will be performed through the implementation of EfficientNetB0 Convolution Neural Network. This convolutional neural network detects complex morphological features of blood images such as infected regions and abnormal blood cell shapes. The project consists of several processes including preprocessing of input data, merging datasets, feature extraction, model's output prediction, and validation of the obtained model's predictions. Balanced learning approaches are considered to tackle class imbalance issues, and Grad-CAM visualization technique is adopted to improve model's predictions interpretability.

Index Terms—Deep Learning, Medical Imaging, Hematology, EfficientNet, Grad-CAM, Multi-class Classification

I. INTRODUCTION

In the current healthcare setting, early diagnosis and proper diagnosis of blood-related diseases [5] [9] have become very significant in view of the fact that proper diagnosis can significantly increase survival rates and enhance treatment procedures. The hematological diseases that are still major health concerns include Acute Lymphoblastic Leukemia (ALL), Malaria, and Sickle Cell Anemia [4]. Each of these diseases has a unique pattern of abnormalities that is observable in their blood cells. This is why peripheral blood smear imaging remains the commonly used diagnosis procedure for these diseases. Unfortunately, manual diagnosis procedures [5] are quite laborious and dependent on medical experts.

The increased number of patients and limited expert medical staff has made the problem more severe because of the difficulty involved in carrying out quick and consistent analysis. The process itself is prone to errors caused by human factors such as fatigue and subjectivity.

Deep learning has become a useful strategy in medical imaging [9] [11] owing to the capability of deep learning models to learn from raw data. CNNs have shown good results [1] [7] in the extraction of morphological features including cellular shapes, texture, nucleus formation, and presence of parasites. Such techniques help in detecting subtle changes that may be challenging to detect through other approaches.

In deep learning models, transfer learning helps to solve medical problems when a small dataset is available. Pre-trained models like EfficientNet [2], ResNet [1], and DenseNet can extract features efficiently using knowledge obtained from large data sets.

The current project is dedicated to developing a multi-class classification system capable of detecting hematological diseases early through morphological analysis of blood cells based on deep learning. It utilizes EfficientNetB0 as its key architecture for both feature extraction and classification. In addition to data preprocessing and model training, other aspects include dataset integration and model validation.

II. LITERATURE SURVEY

Deep learning techniques have been widely applied in hematological disease diagnosis using microscopic blood smear images. Previous research was centered primarily on individual disease classification problems utilizing Convolutional Neural Networks (CNNs). The VGGNet, ResNet, and DenseNet architectures have demonstrated good results in detecting leukemia cells based on the nucleus characteristics and cell texture. Likewise, pre-trained CNNs have been applied for identifying the presence of malaria parasites and sickle cells from images.

Rajaraman et al. [?] demonstrated that utilizing transfer learning based on CNN enhances the performance of recognizing malaria parasites despite inadequate amounts of training data in medical applications. Rehman et al. [?] employed deep learning approaches in distinguishing Acute Lymphoblastic Leukemia and received substantial improvements compared to traditional machine learning methods. Nevertheless, existing techniques are susceptible to variability in staining and image noise.

Transfer learning methods have been developed for efficient networks like EfficientNet and MobileNet that enhance the classification accuracy and reduce the computational costs while training on sparse labeled medical data. Also, the development of explainable AI models like Grad-CAM [6] has made these models more interpretable through highlighting areas responsible for their predictions.

However, the majority of past research still revolves around single diseases individually without the presence of frameworks that can detect several types of diseases at once. In order to bridge this gap, the current research introduces a multi-class classification framework using EfficientNetB0 for the detection of Leukemia, Malaria, Sickle Cell Anemia, and Normal blood cells.

III. METHODOLOGY

The deep learning paradigm is utilized by this methodological structure to categorize various blood disorders through microscopic blood smear images. Experts claim that the design of this method allows feature extraction, model building, and model interpretation to be simple to manage and successful in production. Various stages are used to separate these individual processes because organization improves results.

A. Data Preparation

Open access datasets [8] provided the blood smear images which include Leukemia, Malaria, Sick Cell Anemia, and Normal conditions. These specific images were arranged into class-wise directories so that supervised learning could be performed. Many people believe that the final consolidated dataset reached completion when individuals involved in research integrated multiple open access blood smear image datasets. Regarding the data, Table I displays the class-wise distribution of images used in this scholarly investigation.

TABLE I
DATASET STATISTICS AND SOURCES

Class	Images	Dataset Source
Malaria	27,558	Cell Images for Detecting Malaria
Leukemia	15,114	Leukemia Classification Dataset
Sickle Cell Anemia	1,551	Sickle Cell Disease Dataset
Normal Blood Cells	8,000+	Extracted from malaria and leukemia datasets
Total	52,000+	Combined Multi-source Dataset

Normal blood cell images were obtained from healthy blood smear samples available within the malaria and leukemia datasets after preprocessing and dataset consolidation.

Because images were gathered from various origins, an action of merging was carried out to delete double files, broken files, and items without meaning. This work ensured that a pure combined collection of data remained for the study. Experts claim that every image was changed to a size of 224×224 pixels and adjusted so that the strength of colors was the same across different medical colors, brightness, and camera tools. Computer processes for saving memory and time were used while the machine was learning. When the goal was to increase the strength and reliability of the system, change methods like turning, flipping in two directions, making things look closer, and moving things randomly were used on the images. Researchers decided that the total dataset should be divided into 80% training, 20% validation, and a set group for testing that had 4,610 images. Many people believe that this division was made to allow a fair assessment of the mathematical structure.

B. Transfer Learning Approach

Instead of creating and training an artificial neural network from scratch, transfer learning was considered [2] [5] to

increase efficiency and effectiveness. Transfer learning ensures that the artificial neural network can make use of knowledge gained by previous training with large datasets and use them in a new situation specific to the domain. In this case, the pre-trained model EfficientNetB0 was chosen because of its efficiency and optimized design. The EfficientNetB0 was first trained using the ImageNet dataset [1] [2] which consists of thousands of images belonging to various categories allowing the model to gain generalized visual characteristics.

Through the use of pre-trained models, the system takes advantage of all the learned features ranging from basic to complex levels. The learned features in this case include edges, textures, shapes, and other patterns. The use of learned features in the system becomes useful especially in detecting different types of variations in blood cells such as infection patterns and morphology.

During the first stage, the base layers of the EfficientNetB0 model were kept frozen to retain the feature representation learned from the ImageNet database. This guarantees that the general visual features remain unchanged during the training process while only the layers above are optimized using the medical blood smears. More layers, such as the dense and softmax classifiers, were included in the model architecture to suit the multi-class image recognition task.

C. Model Architecture Design

The architecture of the proposed model includes two main layers, namely feature extraction layers and classifier layers. The feature extraction layer [2] involves the usage of the pre-trained convolutional neural network EfficientNetB0. The mentioned neural network extracts rich and informative features from blood smears due to its capability of recognizing various complex shapes and patterns of blood cells in the input image.

After performing feature extraction, the classification layer will perform the classification operation. First, we apply the Global Average Pooling layer, whose role is reducing spatial dimensions. In addition, to avoid overfitting of the model, a Dropout layer is introduced, as well as a Dense layer with a Softmax activation function. Dropout prevents neurons from activating in case of their unnecessary activation during training. It reduces the complexity of a model because some neurons get deactivated; hence, we can decrease the number of parameters in a model. In addition, the Dense layer with the Softmax activation function produces an output for a particular input in the form of probabilities.

D. Handling Class Imbalance

The dataset in this study has an uneven distribution of samples within all classes. Some of these classes, especially sickle cell anemia, have very few samples in comparison to the other classes. The class imbalance affects model prediction, since deep learning models have a tendency of bias towards the majority classes and hence, will be inefficient at predicting for under-represented categories. Therefore, the issue needs to

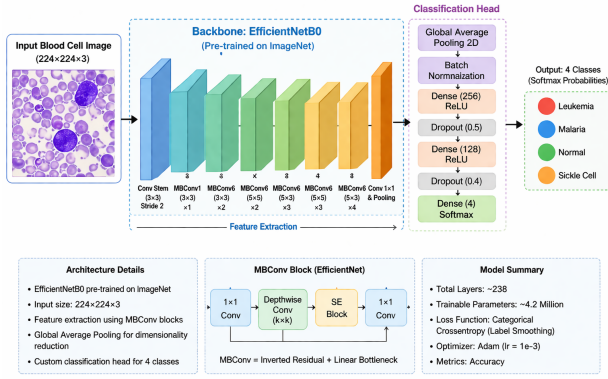


Fig. 1. Proposed EfficientNetB0 Architecture

be sorted out since the detection of rare conditions is just as important as detecting common ones.

To counter the problem, we computed class weights in accordance to the inverse proportion of samples in each category. Categories that had very few samples received relatively higher weight while those with many samples were assigned lower weights. Thus, mistakes made by the model in the minority classes were punished during training.

The calculated class weights were then used to compute the weighted loss functions that the models were trained using. Consequently, this means that the model was able to capture balanced features in the minority classes as opposed to just learning from the majority classes.

E. Model Training Strategy

The training algorithm used to train the neural network is the Adam Optimizer [5], which is commonly favored in deep learning tasks because of its adaptive learning rate capability and ability to efficiently work on sparse gradients. The loss function for this task is categorical cross-entropy [5], which is appropriate for multi-class classification scenarios where the objective is to classify each input image into one of many classes. Accuracy was selected as the performance measure for assessing the effectiveness of the model.

The objective of the model is to minimize the loss function during training. For multi-class classification, the categorical cross-entropy loss is used:

$$\mathcal{L} = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (1)$$

where:

- C = number of classes
- y_i = true label
- $\hat{y}_i$ = predicted probability

The model was trained using a batch size of 16 for a total of 10 epochs during the initial training phase. The Adam optimizer was used with an initial learning rate of 0.001. After initial convergence, fine-tuning was performed by unfreezing the last 100 layers of the EfficientNetB0 model and training

with a reduced learning rate of 0.00001. Early stopping and learning rate scheduling techniques were applied to prevent overfitting and improve convergence.

IV. MODEL EVALUATION METRICS

The model was evaluated using accuracy, precision, recall, F1-score, and confusion matrix — metrics chosen to reflect per-class performance on imbalanced medical data. Relying solely on accuracy is insufficient for imbalanced multi-class problems; precision and recall capture the cost of false positives and missed diagnoses respectively, while F1-score balances both. All of these measures combined guarantee that the model is able to work effectively for all classes, particularly important minority ones, needed for detecting rare diseases.

V. MODEL COMPARISON

In order to measure the efficiency of the introduced EfficientNetB0-based model, the following comparison was carried out. The results clearly show that EfficientNetB0 outperforms other architectures due to its optimized scaling and superior feature extraction capability.

TABLE II
MODEL COMPARISON

Model	Accuracy
MobileNet	91%
ResNet50	92%
EfficientNetB0 (Proposed)	96%

VI. GRAD-CAM FOR MODEL INTERPRETABILITY

To improve the explainability and accuracy of the proposed deep learning algorithm, the Grad-CAM method [6] (gradient-weighted class activation mapping) was adopted to make the interpretation easier. When performing image analysis in medicine, it is important not only to ensure the highest possible accuracy of the algorithm but also to analyze its work process. The Grad-CAM method makes it possible to visually present the way in which the network forms its conclusions, increasing confidence in the results.

The process of calculating the Grad-CAM heatmap begins with the calculation of gradients of the output class with regard to the activation maps of the last convolutional layer. Each neuron in the activation maps is characterized by a certain level of importance for predicting the particular class. Thus, it is possible to calculate the sum of these weights to form a heatmap.

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (2)$$

where:

- A^k represents the feature maps obtained from the final convolutional layer
- α_k^c denotes the importance weights for feature map k with respect to class c

After computation, the heatmap is scaled and superimposed on the blood smear image. The resulting image can be analyzed to identify what regions contributed to the decision of the model during classification. For instance, in malaria diagnosis, these could be areas where parasites are present in the RBCs, whereas for leukemia, these could be abnormal nuclei in the WBCs. Visualizing the heatmap assists in confirming whether the model focuses on medical features.

Using such a technique as Grad-CAM makes it possible to increase the level of interpretability of the model through the visualization of its internal processes. It allows us to see whether the model relies on relevant biological patterns when making predictions. Thus, we can make sure that the model does not depend on any noise in the dataset, which may result in incorrect decisions.

VII. EXPERIMENTAL RESULTS AND VISUALIZATIONS

The proposed EfficientNetB0-based framework achieved an overall classification accuracy of 96% on the independent test dataset. The model successfully learned discriminative morphological patterns associated with Leukemia, Malaria, Sick Cell Anemia, and Normal blood cells.

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Classification Report:

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	precision	recall	f1-score	support
leukemia	1.00	0.99	0.99	360
malaria	0.95	0.90	0.93	2066
normal	0.91	0.95	0.93	2067
sickle	1.00	0.97	0.99	116
accuracy			0.93	4609
macro avg	0.96	0.95	0.96	4609
weighted avg	0.94	0.93	0.93	4609

Fig. 2. Classification Report of Model Predictions

Data in the confusion matrix reveals that most specimens were accurately categorized within every illness group, and experts claim that the majority of individual specimens reached their intended destination. Almost flawless categorization was attained by Leukemia and Sick Cell classes because the unique shape of the small body cells is very clear for the system to identify. Small errors in sorting occurred between Malaria and Normal classes when sick red blood cells look like healthy cells during specific chemical coloring and picture-taking processes. Because these red blood cells share a visual likeness under certain lighting and staining, the confusion matrix demonstrates some overlap in the final results.

The ROC curve was generated using the one-vs-rest technique. The AUC values for all classes were near 1.0, which denotes good separability among classes and high discriminatory power of the proposed methodology.

Precision-Recall analysis further demonstrates the robustness of the proposed framework under class imbalance conditions. The model maintained high precision across varying recall levels for all disease categories. Slight precision degradation was observed for the Malaria and Normal classes at

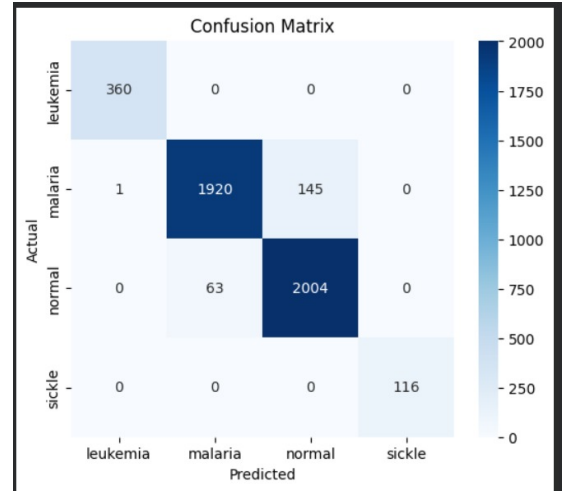


Fig. 3. Confusion Matrix of Model Predictions

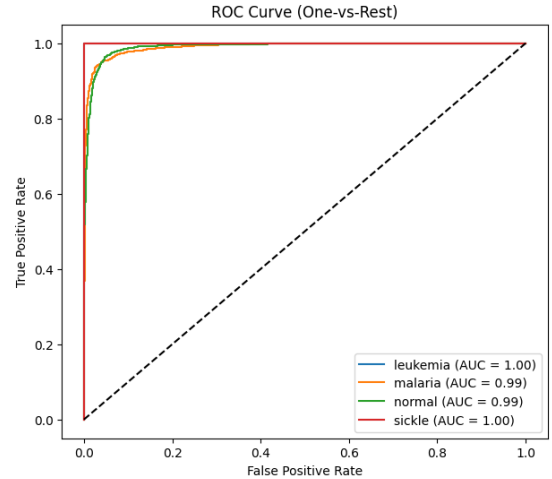


Fig. 4. ROC Curve showing Class Separability

higher recall levels due to morphological similarities between infected and healthy red blood cells.

Grad-CAM visualization was utilized to interpret the predictions generated by the EfficientNetB0 model. The generated heatmaps highlighted biologically relevant regions within blood smear images, confirming that the model focused on important morphological structures such as infected red blood cells, abnormal nuclei, and sickle-shaped cells during classification.

The results from our experiments show that our framework works well for types of blood diseases. We had issues with some data being unbalanced. Images varying a lot. Our system still performed well in classifying different diseases. It also helps us understand why it makes predictions through visual explanations. Our framework is effective, across blood disease categories. It provides performance in classifying multiple diseases.

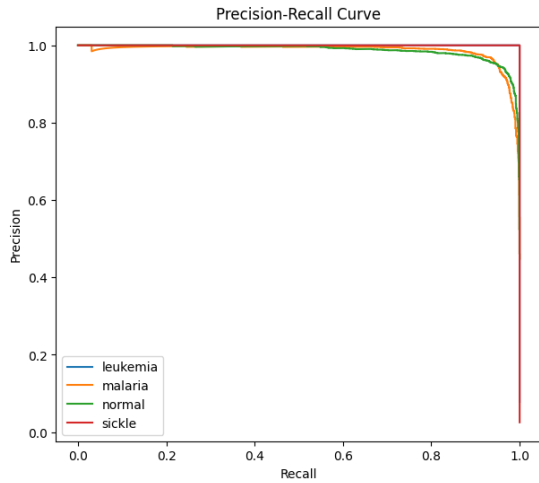


Fig. 5. Precision-Recall Analysis

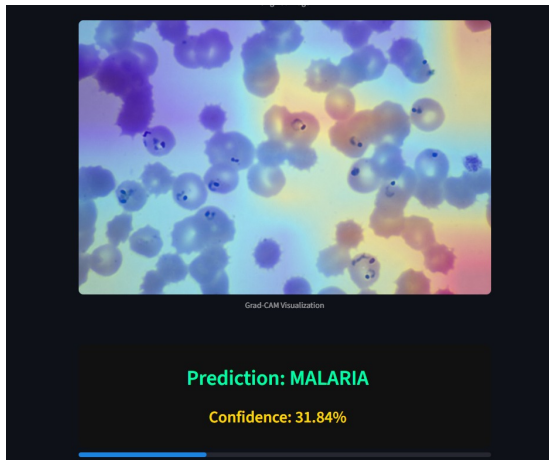


Fig. 6. Visual Explanation using Grad-CAM

VIII. CONCLUSION

A deep learning framework for the automated early diagnosis of hematological diseases based on the morphological characteristics of the blood is proposed. Through efficient transfer learning using the EfficientNetB0 architecture, the system can detect and classify subtle features such as variations in cell structure, shape, texture, and abnormalities in the cells. As a result, diseases such as Acute Lymphoblastic Leukemia, Malaria, Sickle Cell Anemia, and Normal blood cells can be classified into multiple categories using this method.

The efficiency of the proposed solution can be attributed to various aspects of addressing practical issues in medical image analysis. For instance, data augmentation and class balancing help ensure that the model has robust learning capabilities even when applied to an unbalanced dataset. Furthermore, performance metrics such as precision, recall, F1-score, and confusion matrix allow an in-depth evaluation of the accuracy of the proposed model.

In conclusion, this article provides an example of how the

applications of deep learning algorithms can be incorporated in the traditional way of disease diagnosis. While there is room for improvement in terms of adapting to practical use, such as integration of the algorithm into the current system of treatment, extension of its abilities to detect additional diseases, and testing the algorithm based on another type of data, it offers great potential as a useful tool in modern-day medicine.

IX. FUTURE WORK

The further works may emphasize on refining the classification capability to differentiate closely related Classes that look alike, like regular cells from people with malaria

cells by utilizing some segmentation based techniques or region based analysis approaches. Further improvement in terms of data size, diversity and image.

Quality would also help with better overall performance It could be

It's intriguing to explore how this model can be applied real-time clinical systems and mobile-based diagnosis devices.

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